

Report

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Step 1: Data Cleaned Loading

Loading data from CSV File

```
[19] import pandas as pd
      df = pd.read_csv('/content/cleaned_data.csv')
```

Step 2: Data Encryption

Used Caesar Cipher encryption on sensitive text columns in order to maintain confidentiality:

```
[20] def caesar_cipher_encrypt(text, shift=3):
      result = ''
      for char in text:
          if char.isalpha():
              base = ord('A') if char.isupper() else ord('a')
              result += chr((ord(char) - base + shift) % 26 + base)
          else:
              result += char
      return result
```

Encrypted columns: city, region, type.

```
[21] text_columns = ['city', 'region', 'type']
```

Each character was shifted three positions, which was the value used for this operation.

```
[22] for col in text_columns:  
      df[col] = df[col].astype(str).apply(caesar_cipher_encrypt)
```

Step 3: Privacy Policies and Access Control

- Users are assigned roles: admin, user, or guest.

```
def access_data(role):  
    if role == 'admin':  
        return df  
    elif role == 'user':  
        return df[['price', 'area']]  
    else:  
        return "Access Denied"
```

- **Admin:** Full access to all columns.

```
admin_view = access_data('admin')
admin_view.head()
```

	ad_id	type	price	area	bedrooms	bathrooms	level	furnished	rent	city	region
0	Ad id 194281336	dsduwphqw	3790000.0	207.0	3	3	2	NaN	no	jicd	6wk ri rf wrehu
1	Ad id 194198684	dsduwphqw	870000.0	155.0	3	3	4	no	no	fdlur	qhz kholrsrolv
2	Ad id 191102186	dsduwphqw	1546400.0	170.0	3	2	9	no	no	fdlur	cdkudd do pddgl
3	Ad id 193951860	dsduwphqw	1300000.0	270.0	4	3	ground	no	no	vkdufld	10wk ri udpgdgq
4	Ad id 194064010	gxsoha	5888000.0	387.0	4	4	ground	no	no	fdlur	qhz fdlur - ho wdjdprd

- **User:** Limited access to price and area columns only.

```
user_view = access_data('user')
user_view.head()
```

	price	area
0	3790000.0	207.0
1	870000.0	155.0
2	1546400.0	170.0
3	1300000.0	270.0
4	5888000.0	387.0

- **Guest:** Access denied

```
denied_view = access_data('guest')
print(denied_view)
```

Step 4: Data Security Measures

- Hashing (Optional for future work): Should the need arise, numeric IDs could be hashed.

- **Encrypted Data** saved into a new file:

```
[23] df.to_csv('encrypted_data.csv', index=False)
```

Step 5: Compliance with Data Protection Regulations

Sensitive textual attributes (**city, region, type**) were encrypted with the Caesar Cipher algorithm in order to make the data anonymous and undisclosed.

Role-Based Access Control (RBAC) was implemented in order to restrict access to specific fields of data based on the role assigned to the user.

Only authorized users (Admins) can access all columns of data, while normal users only see non-sensitive fields.

These privacy and security controls ensure confidentiality, integrity, and restricted accessibility of the dataset.

The procedures applied are consistent with fundamental data protection principles as provided by laws such as GDPR and CCPA.

ad_id	type	price	area	bedrooms	bathrooms	level	furnished	rent	city	region
Ad id 194281336	dsduwphqw	3790000.0	207.0	3	3	2		no	jlcd	6wk ri rfwrehu
Ad id 194198684	dsduwphqw	870000.0	155.0	3	3	4	no	no	fdlur	qhz kholsroiv
Ad id 191102186	dsduwphqw	1546400.0	170.0	3	2	9	no	no	fdlur	cdkudd do pddgl
Ad id 193951860	dsduwphqw	1300000.0	270.0	4	3	ground	no	no	vkdutld	10wk ri udpdgdq
Ad id 194064010	gxsoha	5888000.0	387.0	4	4	ground	no	no	fdlur	qhz fdlur - ho wdjdprd
Ad id 194021448	dsduwphqw	950000.0	104.0	2	1	7	no	no	fdlur	qdvu flwb
Ad id 194294715	dsduwphqw	2100000.0	160.0	3	2	1	no	no	fdlur	prvwdnedo flwb
Ad id 194320699	dsduwphqw	2550000.0	215.0	3	2	ground	no	no	jlcd	6wk ri rfwrehu
Ad id 194316553	dsduwphqw	3994232.0	160.0	3	3	ground	no	no	fdlur	qhz fdlur - ho wdjdprd
Ad id 194217278	gxsoha	2310000.0	280.0	3	3	ground	no	no	jlcd	6wk ri rfwrehu

Summary:

Sensitive columns encrypted.

Role-based access established and enforced.

Dataset ready for further secure processing and analysis.

Final encrypted dataset and access control policies successfully deployed.